Dawid Tan

Senior Backend Engineer

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Professional Summary

Senior Backend Engineer with 10 years' experience designing and operating production-grade backend platforms across **cloud-native architectures**, **SaaS environments**, and **distributed microservices systems**. Deep expertise building reliable services in **Python**, delivering high-throughput APIs and data processing pipelines that run at scale on **AWS** and **Kubernetes**. Experienced in designing backend systems that ingest and analyze large volumes of web data, combining **Elasticsearch**, **Kafka**, and polyglot data storage to support real-time analytics and investigation workflows. Known for pragmatic architecture decisions, thoughtful trade-offs between scalability and operational simplicity, and strong ownership of production reliability. Regularly mentor engineers, lead technical design discussions, and help teams evolve infrastructure and service boundaries as platforms grow.

Core Skills

Python, FastAPI, AWS, EC2, S3, Lambda, EMR, EKS, Kubernetes, Docker, Microservices Architecture, SaaS Platforms, Elasticsearch, MongoDB, MySQL, Redis, Kafka, Prometheus, Grafana, Kibana, Distributed Systems Design, Cloud Architecture, Backend System Design, High-Performance APIs, Technical Architecture, Code Reviews, Mentorship, Agile Development, Cross-Team Collaboration

Work Experience

Senior Backend Engineer | Focus on Cloud SaaS Platform Architecture

Infosys | Jan 2023 – Present

- Design and implement high-throughput backend services using **Python** and **FastAPI**, supporting SaaS platforms that process large volumes of external web intelligence signals and security indicators.
- Lead architecture discussions for distributed **microservices** running on **AWS** and orchestrated with **Kubernetes**, improving deployment independence and service scalability.
- Built ingestion pipelines integrating **Kafka**, **MongoDB**, and **Elasticsearch** to collect and index large datasets used for security analytics and investigative workflows.
- Improved API responsiveness by refactoring blocking operations into asynchronous **FastAPI** services, reducing average response latency by 35%.
- Implemented scalable infrastructure patterns across **AWS EC2**, **S3**, and **Lambda** to support variable workloads and automated data processing jobs.

- Introduced platform-wide observability using **Prometheus**, **Grafana**, and **Kibana**, helping teams identify performance regressions and reduce mean time to resolution during production incidents.
- Collaborated with data and security engineers to design backend services capable of analyzing signals from multiple parts of the web ecosystem.
- Conduct **code reviews** and **architecture reviews**, helping backend developers improve service boundaries, maintainability, and long-term scalability of **Python** services.
- Mentor engineers on production debugging, distributed system design, and effective use of **Kubernetes** and **AWS** infrastructure.
- Contribute to ongoing improvements of the engineering stack, evaluating tools and infrastructure patterns that support the platform's growing scale.

Main Stack: Python, FastAPI, AWS, EC2, S3, Lambda, EMR, EKS, Kubernetes, Docker, Microservices, Kafka, Elasticsearch, MongoDB, MySQL, Redis, Prometheus, Grafana, Kibana, SaaS platforms, distributed backend systems, Agile

Backend Engineer | Focus on Scalable E-commerce Platform Services

Sendo.vn | Jan 2021 – Dec 2022

- Developed backend services using **Python** supporting high-traffic **SaaS web applications** used by millions of marketplace users.
- Designed containerized services deployed through **Docker** and orchestrated with **Kubernetes**, enabling reliable scaling during seasonal traffic spikes.
- Implemented event-driven processing using **Kafka**, allowing multiple services to react to order and inventory events without tight coupling.
- Built and optimized search capabilities powered by **Elasticsearch**, significantly improving product discovery performance.
- Tuned data access patterns across **MySQL**, **Redis**, and **MongoDB** to reduce database contention during peak platform usage.
- Migrated legacy synchronous APIs to **FastAPI**, improving developer productivity and increasing API throughput.
- Introduced monitoring dashboards using **Prometheus** and **Grafana**, giving engineering teams clearer insight into system health and performance.
- Partnered with platform engineers to refine **AWS** deployment patterns and reduce operational friction for new services.

Main Stack: Python, FastAPI, AWS, Docker, Kubernetes, Microservices, Kafka, Elasticsearch, MongoDB, MySQL, Redis, Prometheus, Grafana, Kibana, SaaS backend platforms, Agile

Backend Engineer | Focus on Distributed Backend Systems

Global CyberSoft | Jul 2018 – Dec 2020

- Built backend APIs and services using **Python** for distributed web applications deployed on **AWS** infrastructure.

- Designed modular backend components that improved maintainability and simplified long-term system evolution.
- Containerized several legacy services using **Docker**, enabling more consistent deployments and laying groundwork for future microservice adoption.
- Implemented data services integrating **MongoDB**, **MySQL**, and **Redis** to support transactional workflows, caching layers, and flexible document storage.
- Contributed to search and analytics functionality using **Elasticsearch**, supporting indexing and retrieval of large application datasets.
- Investigated and resolved production issues across distributed **Python** services, improving reliability of critical backend systems.
- Participated in **Agile** development cycles, contributing to system design discussions and iterative architecture improvements.

Main Stack: Python, AWS, Docker, Elasticsearch, MongoDB, MySQL, Redis, SaaS backend systems, distributed services, Agile

Education

Hanoi University of Science and Technology | 2013- 2018

Bachelor's Degree of Computer Science